

Multistage variance estimation with `qvar_ms`

Thomas Delclite

2026-09-02

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1 Introduction

The `qvar()` function performs analytical variance estimation in the most common single-stage case: stratified simple random sampling, non-response correction through reweighting, calibration on margins. `qvar_ms()` extends this logic to **multistage designs**: units of stage 2 are drawn within the units of stage 1, units of stage 3 within the units of stage 2, and so on, with non-response correction possibly occurring at each stage and calibration at one given stage.

The variance is estimated as the sum of the conditional variances of each stage, following the Rao (1975) formula as implemented in the SAS Poulpe macros (Caron, 1998) and natively supported by the variance functions of the `gustave` package: at each stage, the variance contributions (`var_srs()`, `var_pois()`) are *linearly* weighted by the cumulated corrected weight of the stages above, through their `w` argument, and the first-stage variance is computed on the estimated totals per primary sampling unit.

This vignette walks through **nine cases of increasing complexity**, using the simulated Labour Force Survey (LFS) data shipped with the package (`lfs_samp_area`: areas, i.e. primary sampling units; `lfs_samp_dwel`: dwellings sampled within areas; `lfs_samp_ind`: individuals within dwellings).

```
library(gustave)
library(dplyr)
library(purrr)

lfs_samp_area <- gustave::lfs_samp_area
lfs_samp_dwel <- gustave::lfs_samp_dwel
lfs_samp_ind  <- gustave::lfs_samp_ind
```

2 How to use `qvar_ms()`

The best way to learn `qvar_ms()` is to build wrappers of increasing complexity. This section walks through nine cases, each introducing exactly one new feature of the interface: from a single-stage design that reproduces `qvar()` (case 1), through two-stage designs with non-response correction, stratification and calibration (cases 2 to 6), to three-stage designs with corrections at several stages and calibration at an intermediate stage (cases 7 to 9).

The cases are cumulative — the data prepared in one case is reused in the next — and follow a fixed pattern: the `qvar_ms()` call first, then a hand-built equivalent using `define_variance_wrapper()` and the core `gustave` functions, which makes every methodological step explicit and serves as ground truth. Readers in a hurry can jump to the case matching their design and work backwards.

`qvar_ms()` uses `varDT()` to estimate the variance of the estimator of a total in the case of a balanced sampling design with equal or unequal probabilities using Deville-Tillé (2005) formula. All examples use the function `var_srs()`, which is a convenience wrapper for the (stratified) simple random sampling case.

2.1 Conventions

Before anything else, the conventions used by `qvar_ms()` — most errors in practice come from overlooking one of them:

- **Stage ordering.** Stage 1 is the highest sampling stage (e.g. primary sampling units); stage L = `sampling_stages` is the final stage (surveyed units). `data` is a list of L `data.frames` ordered from stage 1 to stage L , each containing **all** the units sampled at the corresponding stage, including out-of-scope and non-responding units.
- **Parent linkage.** Each file from stage 2 onwards contains a variable identifying the unit of the stage above (`parent_id` argument; by default, the `id` variable of the previous stage is assumed to be present).
- **Conditional weights.** `sampling_weight` and `nrc_weight` are **conditional** weights at the considered stage: the inverse of the inclusion (resp. response) probability *given the stages above*. This is the single most important convention of the interface.
- **Response probability.** `response_prob` can be used instead of `nrc_weight` to include response probability at each stage of non-response.
- **Cumulated calibration weight.** `calibration_weight` is the **cumulated** weight at the calibration stage — the weight actually used in the calibration equations.
- **Last stage only.** `dissemination_dummy`, `dissemination_weight` and `scope_dummy` refer to the last stage file.
- **Per-stage arguments.** Non-response correction may be declared at each stage through list-valued arguments (`nrc_weight`, `response_dummy`, `nrc_dummy`); a single variable name is assigned to the last stage. Calibration is declared at one single stage (`calibration_stage`, last stage by default).

Argument	Per stage?	Weight convention	Default
<code>id</code>	yes (length L)	—	required
<code>parent_id</code>	stages 2.. L	—	id of stage above
<code>sampling_weight</code>	yes (length L)	conditional	required
<code>strata</code>	yes (list)	—	none
<code>nrc_weight</code> ,	yes (list)	conditional	last stage
<code>response_dummy</code> ,			
<code>response_dummy</code> ,			
<code>nrc_dummy</code>			
<code>calibration_stage</code>	one stage	—	last stage
<code>calibration_weight</code>	one stage	cumulated	none
<code>dissemination_*</code> ,	last stage	cumulated	required / none
<code>scope_dummy</code>			

2.2 Case 1: single stage — backward compatibility with `qvar()`

When `data` is a single `data.frame` (or `sampling_stages = 1`), `qvar_ms()` degenerates to the standard single-stage case and should reproduce `qvar()` exactly. This is the first thing to check.

```
lfs_samp_dwel$diss    <- TRUE
lfs_samp_dwel$w_dwel <- 1 / lfs_samp_dwel$pik_dwel
lfs_samp_dwel$w_final <- 1 / lfs_samp_dwel$pik
```

```
qvar1 <- qvar_ms(
  data = lfs_samp_dwel,
  id = "id_dwel",
```

```

dissemination_dummy = "diss",
dissemination_weight = "w_final",
sampling_weight = "w_final",
define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - simple random sampling WITHOUT stratification
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

qvar1b <- qvar(
  data = lfs_samp_dwel,
  id = "id_dwel",
  dissemination_dummy = "diss",
  dissemination_weight = "w_final",
  sampling_weight = "w_final",
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - simple random sampling WITHOUT stratification
#>
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

qvar1(lfs_samp_dwel, total(income))
#>           call n      est variance      std      cv lower upper
#> 1 total(y = income) 80 198417.3 136946649 11702.42 5.897882 175481 221353.7
qvar1b(lfs_samp_dwel, total(income))
#>           call n      est variance      std      cv lower upper
#> 1 total(y = income) 80 198417.3 136946649 11702.42 5.897882 175481 221353.7

```

Both wrappers return the same estimate and the same variance: with one stage, the cumulated weight of the “stages above” is 1 everywhere and the Rao weighting is inactive.

2.3 Case 2: two-stage design, no non-response, no calibration

We now use the area file as a first stage. For this introductory case the area weights are made equal (no stratification at stage 1), as `qvar_ms()` — like `qvar()` — relies on the simple random sampling formula within strata and would otherwise warn and use the mean weight per stratum. The final dwelling weight is rebuilt as the product of the **conditional** weights of the two stages.

```

lfs_samp_area$w_area <- base::mean(1 / lfs_samp_area$pik_area)

lfs_samp_dwel <- lfs_samp_dwel %>%
  left_join(lfs_samp_area %>% select(id_area, w_area), by = "id_area")

lfs_samp_dwel$w_dwel <- 1 / lfs_samp_dwel$pik_dwel
lfs_samp_dwel$w_final <- lfs_samp_dwel$w_dwel * lfs_samp_dwel$w_area

```

```

qvar2 <- qvar_ms(
  # One file per stage, from the highest to the lowest
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = c("id_area"),
  dissemination_dummy = "diss",
  dissemination_weight = "w_final",
  # Conditional sampling weights, one per stage
  sampling_weight = c("w_area", "w_dwel"),
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 2-stage sampling design
#>   - stage 1: simple random sampling WITHOUT stratification
#>   - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↪ stage above)
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↪ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

```

The hand-built equivalent makes the Rao (1975) decomposition explicit: one within-area variance per primary unit, multiplied by the area weight (*first power* — this is the Rao row weight), plus the first-stage variance computed on the estimated area totals. The generalization to more stages and to non-response phases proceeds recursively, the Rao row weight at stage k being the cumulated corrected weight of stages $1, \dots, k-1$. This is the estimator implemented by the SAS Poulpe macros (Caron, 1998).

```

var_srs_two <- function(y, smp_area, smp_dwel) {

  variance <- list()
  y <- add_zero(y, rownames = smp_dwel$id_dwel)

  # Stage 2: one SRS variance per area, weighted by the area weight
  variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
    sub <- smp_dwel %>% filter(id_area == .x)
    w_area <- unique(sub$w_area)
    var_srs(
      y[sub$id_dwel, , drop = FALSE],
      pik = sub$pik_dwel
    ) * w_area
  }) %>% unlist %>% sum

  # Aggregation: estimated totals per area
  y_area <- sum_by(y, by = smp_dwel$id_area, w = 1 / smp_dwel$pik_dwel)

  # Stage 1: SRS variance on the estimated area totals
  variance[["srs-area"]] <- var_srs(
    y_area[smp_area$id_area, , drop = FALSE],
    pik = 1 / smp_area$w_area
  )

  print(as.data.frame(variance))
}

```

```

  Reduce(`+`, variance)
}

qvar2b <- define_variance_wrapper(
  variance_function = var_srs_two,
  technical_data    = list(smp_area = lfs_samp_area, smp_dwel = lfs_samp_dwel),
  reference_id      = lfs_samp_dwel$id_dwel,
  reference_weight   = lfs_samp_dwel$w_final,
  default_id        = "id_dwel"
)

qvar2(lfs_samp_dwel, total(income))
#>           call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 198343.6 523055781 22870.41 11.5307 153518.4 243168.8
qvar2b(lfs_samp_dwel, total(income))
#>   srs.dwel srs.area
#> 1 5096232 517959549
#>           call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 198343.6 523055781 22870.41 11.5307 153518.4 243168.8

```

Note how `qvar_ms()` achieves the per-area computation: at stage 2, the effective strata are the crossing of the parent unit and of the declared strata variable (here, none), so that one single stratified `var_srs()` call is algebraically identical to the sum of per-area variances — the Rao weight, constant within each area, being passed through the `w` argument. The components printed by the hand-built version (`srs-dwel`, `srs-area`) correspond one-to-one to the `sampling_stage2` and `sampling_stage1` components that `qvar_ms()` reports in its “components” diagnostic (see diagnostic section below):

```

qvar2(lfs_samp_dwel, total(income), diagnostics = "components")
#>
#> -- Variance components --
#>   sampling_stage2      5.096e+06      1.0 %
#>   sampling_stage1      5.18e+08     99.0 % #####
#>   TOTAL                5.231e+08
#>           call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 198343.6 523055781 22870.41 11.5307 153518.4 243168.8

```

2.4 Case 3: non-response correction at the last stage

We simulate a response mechanism at the dwelling stage. **The non-response correction weight passed to `qvar_ms()` must be conditional to the current stage:** `w_nrc = w_dwel / prep`, not `w_final / prep`. Passing a cumulated weight here is the most common mistake — it silently deflates the response probabilities computed internally (`sampling_weight / nrc_weight`) and propagates a spurious factor to every stage of the decomposition. A safer alternative is to pass the estimated response probabilities directly, through the `response_prob` argument: probabilities have no cumulated counterpart, so this mistake simply cannot occur. The two arguments are mutually exclusive — declare one or the other at each corrected stage, not both.

```

set.seed(123)
lfs_samp_dwel$rand <- runif(nrow(lfs_samp_dwel))
lfs_samp_dwel$rep  <- lfs_samp_dwel$rand <= 0.6
lfs_samp_dwel$prep  <- 0.6
lfs_samp_dwel$w_nrc <- lfs_samp_dwel$w_dwel / lfs_samp_dwel$prep # conditional!

```

```

lfs_samp_dwel$w_final <- lfs_samp_dwel$w_nrc * lfs_samp_dwel$w_area
lfs_samp_dwel$w_final[!lfs_samp_dwel$rep] <- 0
lfs_samp_dwel$income[!lfs_samp_dwel$rep] <- 0

```

```

qvar3 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = c("id_area"),
  dissemination_dummy = "rep",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel"),
  # Single names: assigned to the last stage by default
  nrc_weight = "w_nrc",
  # response_prob = "prep", # instead of nrc_weight
  response_dummy = "rep",
  define = TRUE
)
#> Note: The nrc_weight argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - 2-stage sampling design
#> - stage 1: simple random sampling WITHOUT stratification
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

```

In the hand-built version, the non-response variance is a Poisson term whose Rao row weight is the cumulated design weight down to the current stage ($w_{dwel} * w_{area}$), and the interest variable is expanded by the inverse response probability before moving up:

```

var_srs_two <- function(y, smp_area, smp_dwel) {

  variance <- list()
  resp_dwel <- smp_dwel %>% filter(rep == 1)

  # Non-response variance (Poisson, respondents only)
  variance[["nr"]] <- var_pois(
    y[resp_dwel$id_dwel, , drop = FALSE],
    pik = resp_dwel$prep,
    w = resp_dwel$w_dwel * resp_dwel$w_area
  )

  # Reintroduce non-responding dwellings with zeros
  y <- add_zero(y, rownames = smp_dwel$id_dwel)

  # Stage 2 sampling variance on the NR-expanded variable, over ALL

```

```

# sampled dwellings (non-respondents count as zeros)
variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
  sub <- smp_dwel %>% filter(id_area == .x)
  w_area <- unique(sub$w_area)
  var_srs(
    y[sub$id_dwel, , drop = FALSE] / sub$prep,
    pik = sub$pik_dwel
  ) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

variance[["srs-area"]] <- var_srs(
  y_area[smp_area$id_area, , drop = FALSE],
  pik = 1 / smp_area$w_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar3b <- define_variance_wrapper(
  variance_function = var_srs_two,
  technical_data     = list(smp_area = lfs_samp_area, smp_dwel = lfs_samp_dwel),
  reference_id       = lfs_samp_dwel$id_dwel,
  reference_weight    = lfs_samp_dwel$w_final,
  default_id         = "id_dwel"
)

qvar3(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   discarded.
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 352284542 18769.24 9.855252 153662.1 227236.2
qvar3b(lfs_samp_dwel, total(income))
#>           nr srs.dwel  srs.area
#> 1 3213255 22003800 327067487
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 352284542 18769.24 9.855252 153662.1 227236.2

```

Two things worth noticing. First, the second-stage sampling variance is computed over **all sampled dwellings**, respondents and non-respondents alike (the latter with zeros): this is why each stage file must contain all sampled units. Second, `add_zero()` occurs at the top of each stage block inside `qvar_ms()`: it does not include non-respondents in the calibration or non-response steps (those explicitly select their rows), it only guarantees their presence — with a zero — at the sampling variance step. Zeroed non-respondents are not a computational trick but the exact HT computation — see the appendix.

2.5 Case 4: stratified first stage with a single-unit stratum

We now stratify the first stage, deliberately putting a single area in stratum A. Like `qvar()`, `qvar_ms()` warns, excludes such strata from the variance estimation and keeps them for point estimates. It is also possible to estimate variance of single-unit stratum with a Poisson variance term, with the argument `single_unit_strata = "poisson"`.


```
lfs_samp_area$strata <- ifelse(lfs_samp_area$id_area == "A016", "A", "B")
```

```
qvar4 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = c("id_area"),
  dissemination_dummy = "rep",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel"),
  # One element per stage: strata at stage 1, none at stage 2
  strata = list("strata", NULL),
  nrc_weight = "w_nrc",
  response_dummy = "rep",
  define = TRUE
)
#> Note: The nrc_weight argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - 2-stage sampling design
#> - stage 1: stratified simple random sampling
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> Note: The strata variable (strata) at stage 1 is of type character. It is
  ↳ automatically coerced to factor.
#> Warning: The following strata at stage 1 contain less than two sampled units: A. They
  ↳ are excluded from the variance estimation process (but kept for point estimates).
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

The hand-built version has to remove the single-unit strata explicitly before calling `var_srs()` (which would otherwise stop), reproducing the behaviour of `qvar_ms()`:

```
var_srs_two <- function(y, smp_area, smp_dwel) {
  variance <- list()
  resp_dwel <- smp_dwel %>% filter(rep == 1)

  variance[["nr"]] <- var_pois(
    y[resp_dwel$id_dwel, , drop = FALSE],
    pik = resp_dwel$prep,
    w = resp_dwel$w_dwel * resp_dwel$w_area
  )

  y <- add_zero(y, rownames = smp_dwel$id_dwel)

  variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
    sub <- smp_dwel %>% filter(id_area == .x)
```

```

w_area <- unique(sub$w_area)
var_srs(
  y[sub$id_dwel, , drop = FALSE] / sub$prep,
  pik = sub$pik_dwel
) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

# Exclude single-unit strata (kept for point estimates)
table_strata <- as.data.frame(table(smp_area$strata))
single_unit <- table_strata[which(table_strata$Freq == 1), 1]
smp_area_ok <- smp_area %>% filter(!strata %in% single_unit)

variance[["srs-area"]] <- var_srs(
  y_area[smp_area_ok$id_area, , drop = FALSE],
  pik = 1 / smp_area_ok$w_area,
  strata = smp_area_ok$strata
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar4b <- define_variance_wrapper(
  variance_function = var_srs_two,
  technical_data = list(smp_area = lfs_samp_area, smp_dwel = lfs_samp_dwel),
  reference_id = lfs_samp_dwel$id_dwel,
  reference_weight = lfs_samp_dwel$w_final,
  default_id = "id_dwel"
)

qvar4(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
  ↳ discarded.
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 128144632 11320.1 5.943895 168262.2 212636.1
qvar4(lfs_samp_dwel, total(income), single_unit_strata = "poisson")
#> Warning: 32 observations do not match any responding units of the survey. They are
  ↳ discarded.
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 3482048661 59008.89 30.98406 74793.86 306104.4
qvar4b(lfs_samp_dwel, total(income))
#>      nr srs.dwel srs.area
#> 1 3213255 22003800 102927577
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 128144632 11320.1 5.943895 168262.2 212636.1

```

2.6 Case 5: stratification at both stages

Declared strata at stage 2 or beyond are automatically **crossed with the parent unit**, since units are drawn within the units of the stage above: declaring `strata_dwel` at stage 2 means stratified simple random sampling of dwellings *within each area*. The hand-built version makes this explicit by passing `strata =`

sub\$strata_dwel inside each per-area call.

```
lfs_samp_area$strata_area <- ifelse(lfs_samp_area$id_area == "A016", "A", "B")
lfs_samp_dwel$strata_dwel <- paste0("S", as.numeric(as.factor(lfs_samp_dwel$id_dwel))) %%
  3)
```

```
qvar5 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = c("id_area"),
  dissemination_dummy = "rep",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel"),
  strata = list("strata_area", "strata_dwel"),
  nrc_weight = "w_nrc",
  response_dummy = "rep",
  define = TRUE
)
#> Note: The nrc_weight argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 2-stage sampling design
#>   - stage 1: stratified simple random sampling
#>   - stage 2: stratified simple random sampling (within the units of the stage above),
  ↳ with non-response correction through reweighting
#> Note: The strata variable (strata_area) at stage 1 is of type character. It is
  ↳ automatically coerced to factor.
#> Note: The strata variable (strata_dwel) at stage 2 is of type character. It is
  ↳ automatically coerced to factor.
#> Warning: The following strata at stage 1 contain less than two sampled units: A. They
  ↳ are excluded from the variance estimation process (but kept for point estimates).
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area) and
  ↳ of the strata variable (strata_dwel).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

```
var_srs_two <- function(y, smp_area, smp_dwel) {

  variance <- list()
  resp_dwel <- smp_dwel %>% filter(rep == 1)

  variance[["nr"]] <- var_pois(
    y[resp_dwel$id_dwel, , drop = FALSE],
    pik = resp_dwel$prep,
    w = resp_dwel$w_dwel * resp_dwel$w_area
  )

  y <- add_zero(y, rownames = smp_dwel$id_dwel)
```

```

# Stage 2: stratified SRS within each area (parent x declared strata)
variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
  sub <- smp_dwel %>% filter(id_area == .x)
  w_area <- unique(sub$w_area)
  var_srs(
    y[sub$id_dwel, , drop = FALSE] / sub$prep,
    pik = sub$pik_dwel,
    strata = sub$strata_dwel
  ) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

table_strata <- as.data.frame(table(smp_area$strata_area))
single_unit <- table_strata[which(table_strata$Freq == 1), 1]
smp_area_ok <- smp_area %>% filter(!strata_area %in% single_unit)

variance[["srs-area"]] <- var_srs(
  y_area[smp_area_ok$id_area, , drop = FALSE],
  pik = 1 / smp_area_ok$w_area,
  strata = smp_area_ok$strata_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar5b <- define_variance_wrapper(
  variance_function = var_srs_two,
  technical_data = list(smp_area = lfs_samp_area, smp_dwel = lfs_samp_dwel),
  reference_id = lfs_samp_dwel$id_dwel,
  reference_weight = lfs_samp_dwel$w_final,
  default_id = "id_dwel"
)

qvar5(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
↪ discarded.
#>      call      n      est variance      std      cv lower upper
#> 1 total(y = income) 48 190449.2 128922218 11354.39 5.961902 168195 212703.4
qvar5b(lfs_samp_dwel, total(income))
#>      nr srs.dwel srs.area
#> 1 3213255 22781386 102927577
#>      call      n      est variance      std      cv lower upper
#> 1 total(y = income) 80 190449.2 128922218 11354.39 5.961902 168195 212703.4

```

Note that with strata crossed with parent units, single-unit *crossed* strata (an area containing a single sampled dwelling in a given stratum) are excluded by default with a warning, just as at stage 1.

2.7 Case 6: calibration at the last stage

Calibration enters the variance estimation through the residual technique (`res_cal()`): the interest variable is replaced by its residuals on the calibration variables before the rest of the decomposition. Two conventions matter here:

- `calibration_weight` is the **cumulated** weight at the calibration stage — the weight actually used in the calibration equations (here `w_final`);
- qualitative calibration variables (character or factor) are automatically discretized into full sets of dummies. The hand-built version uses `model.matrix()` (intercept + treatment contrasts) instead.

```
qvar6 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = c("id_area"),
  dissemination_dummy = "rep",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel"),
  strata = list("strata_area", NULL),
  nrc_weight = "w_nrc",
  response_dummy = "rep",
  # Calibration at the last stage (default), CUMULATED weight
  calibration_weight = "w_final",
  calibration_var = "strata_dwel",
  define = TRUE
)

#> Note: The nrc_weight argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 2-stage sampling design
#>   - stage 1: stratified simple random sampling
#>   - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#>   - calibration on margins at stage 2
#> Note: The strata variable (strata_area) at stage 1 is of type character. It is
  ↳ automatically coerced to factor.
#> Note: The following calibration variables are qualitative (factor, character):
  ↳ strata_dwel. They will be automatically discretized.
#> Warning: The following strata at stage 1 contain less than two sampled units: A. They
  ↳ are excluded from the variance estimation process (but kept for point estimates).
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

```
var_srs_two <- function(y, smp_area, smp_dwel) {

  variance <- list()
  resp_dwel <- smp_dwel %>% filter(rep == 1)
```

```

# Calibration residuals (cumulated weights)
x_cal <- model.matrix(~ strata_dwel, data = resp_dwel)
y[resp_dwel$id_dwel, ] <- res_cal(
  y[resp_dwel$id_dwel, ],
  x = x_cal,
  w = resp_dwel$w_final
)

variance[["nr"]] <- var_pois(
  y[resp_dwel$id_dwel, , drop = FALSE],
  pik = resp_dwel$prep,
  w = resp_dwel$w_dwel * resp_dwel$w_area
)

y <- add_zero(y, rownames = smp_dwel$id_dwel)

variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
  sub <- smp_dwel %>% filter(id_area == .x)
  w_area <- unique(sub$w_area)
  var_srs(
    y[sub$id_dwel, , drop = FALSE] / sub$prep,
    pik = sub$pik_dwel
  ) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

table_strata <- as.data.frame(table(smp_area$strata_area))
single_unit <- table_strata[which(table_strata$Freq == 1), 1]
smp_area_ok <- smp_area %>% filter(!strata_area %in% single_unit)

variance[["srs-area"]] <- var_srs(
  y_area[smp_area_ok$id_area, , drop = FALSE],
  pik = 1 / smp_area_ok$w_area,
  strata = smp_area_ok$strata_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar6b <- define_variance_wrapper(
  variance_function = var_srs_two,
  technical_data = list(smp_area = lfs_samp_area, smp_dwel = lfs_samp_dwel),
  reference_id = lfs_samp_dwel$id_dwel,
  reference_weight = lfs_samp_dwel$w_final,
  default_id = "id_dwel"
)

qvar6(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   discarded.
#>      call      n      est variance      std      cv      lower upper

```

```
#> 1 total(y = income) 48 190449.2 248325893 15758.36 8.274313 159563.3 221335
qvar6b(lfs_samp_dwel, total(income))
#>      nr srs.dwel srs.area
#> 1 659221 8082131 239584540
#>      call n      est variance      std      cv      lower upper
#> 1 total(y = income) 80 190449.2 248325893 15758.36 8.274313 159563.3 221335
```

`qvar_ms()` also compares, at the calibration stage, the calibrated weights with the cumulated weights guessed from the survey description: their ratio is the calibration g-weight and should remain close to 1. A systematically large ratio triggers a warning, and most often reveals that some sampling or non-response weight is cumulated whereas a conditional weight is expected (see case 3).

2.8 Case 7: three stages, exhaustive final stage

We add the individual stage. In the LFS design, all individuals of a responding dwelling are surveyed: the third stage is **exhaustive** (`pik_ind = 1`). Two consequences: `varDT()` internally discards exhaustive units (with a note), so the third-stage sampling variance component is zero; and since the parent stage carries non-response, the individual file must only contain individuals of **responding** dwellings — `qvar_ms()` checks that no unit is attached to a non-responding parent.

```
lfs_samp_ind <- lfs_samp_ind %>%
  left_join(
    lfs_samp_dwel %>% select(id_dwel, rep, pik_dwel, w_dwel, w_final),
    by = "id_dwel"
  ) %>%
  filter(rep) %>%
  mutate(w_ind = 1, pik_ind = 1)

head(lfs_samp_ind)
#>   id_ind id_dwel  income unemp sampling_weight rep pik_dwel w_dwel w_final
#> 1 I00128 D05448 10.990279 FALSE      120.2375 TRUE  0.1666667  6.00 199.1537
#> 2 I00416 D05651  6.844813 FALSE      120.2375 TRUE  0.1652893  6.05 200.8133
#> 3 I01216 D00784 11.602062  TRUE      120.2375 TRUE  0.1652893  6.05 200.8133
#> 4 I01374 D00012 10.712888 FALSE      120.2375 TRUE  0.1666667  6.00 199.1537
#> 5 I01483 D00387  3.485693 FALSE      120.2375 TRUE  0.1652893  6.05 200.8133
#> 6 I01514 D04784 10.989525 FALSE      120.2375 TRUE  0.1652893  6.05 200.8133
#>   w_ind pik_ind
#> 1     1      1
#> 2     1      1
#> 3     1      1
#> 4     1      1
#> 5     1      1
#> 6     1      1
```

```
qvar7 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind),
  sampling_stages = 3,
  id = c("id_area", "id_dwel", "id_ind"),
  parent_id = c("id_area", "id_dwel"),
  dissemination_dummy = "rep",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel", "w_ind"),
```

```

# List syntax: NR correction at stage 2 only
nrc_weight = list(NULL, "w_nrc", NULL),
response_dummy = list(NULL, "rep", NULL),
define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - 3-stage sampling design
#> - stage 1: simple random sampling WITHOUT stratification
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> - stage 3: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above)
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: At stage 3, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_dwel).
#> Warning: The following strata at stage 3 contain less than two sampled units:
  ↳ D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1,
  ↳ D02653:1, D03338:1 and 18 more. They are excluded from the variance estimation
  ↳ process (but kept for point estimates).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

```

```

var_srs_three <- function(y, smp_area, smp_dwel, smp_ind) {

  variance <- list()

  # Stage 3 (individuals): exhaustive, variance contribution is zero
  y <- add_zero(y, rownames = smp_ind$id_ind)
  y_dwel <- sum_by(y, by = smp_ind$id_dwel, w = smp_ind$w_ind)

  # Stage 2 (dwellings): NR + sampling
  resp_dwel <- smp_dwel %>% filter(rep == 1)

  variance[["nr"]] <- var_pois(
    y_dwel[resp_dwel$id_dwel, , drop = FALSE],
    pik = resp_dwel$prep,
    w = resp_dwel$w_dwel * resp_dwel$w_area
  )

  y_dwel <- add_zero(y_dwel, rownames = smp_dwel$id_dwel)

  variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
    sub <- smp_dwel %>% filter(id_area == .x)
    w_area <- unique(sub$w_area)
    var_srs(
      y_dwel[sub$id_dwel, , drop = FALSE] / sub$prep,
      pik = sub$pik_dwel
    ) * w_area
  }) %>% unlist %>% sum

  y_area <- sum_by(y_dwel / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

```



```

# Stage 1 (areas)
variance[["srs-area"]] <- var_srs(
  y_area[smp_area$id_area, , drop = FALSE],
  pik = 1 / smp_area$w_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar7b <- define_variance_wrapper(
  variance_function = var_srs_three,
  technical_data    = list(smp_area = lfs_samp_area,
                           smp_dwel = lfs_samp_dwel,
                           smp_ind  = lfs_samp_ind),
  reference_id      = lfs_samp_ind$id_ind,
  reference_weight   = lfs_samp_ind$w_final,
  default_id        = "id_ind"
)

qvar7(lfs_samp_ind, total(income))
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2 352284542 18769.24 9.855252 153662.1 227236.2
qvar7b(lfs_samp_ind, total(income))
#>      nr srs.dwel srs.area
#> 1 3213255 22003800 327067487
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2 352284542 18769.24 9.855252 153662.1 227236.2

```

2.9 Case 8: non-response correction at two stages

Non-response may occur at several stages — here, total dwelling non-response (a whole cluster of individuals is lost) *and* individual non-response within responding dwellings. The list-valued arguments make the declaration explicit, one element per stage. Both `w_nrc` weights are, as always, **conditional** to their stage.

```

lfs_samp_ind <- gustave::lfs_samp_ind %>%
  left_join(
    lfs_samp_dwel %>% select(id_dwel, rep_hh = rep, pik_dwel,
                           w_nrc_dwel = w_nrc, w_area, w_dwel, w_final),
    by = "id_dwel"
  ) %>%
  filter(rep_hh) %>%
  mutate(w_ind = 1, pik_ind = 1)

set.seed(123)
lfs_samp_ind$rand_ind <- runif(nrow(lfs_samp_ind))
lfs_samp_ind$rep_ind  <- lfs_samp_ind$rand_ind <= 0.8
lfs_samp_ind$prep_ind <- 0.8
lfs_samp_ind$w_nrc    <- lfs_samp_ind$w_ind / lfs_samp_ind$prep_ind # conditional!
lfs_samp_ind$w_final  <- lfs_samp_ind$w_area * lfs_samp_ind$w_nrc_dwel *
  ↪ lfs_samp_ind$w_nrc
lfs_samp_ind$w_final[!lfs_samp_ind$rep_ind] <- 0
lfs_samp_ind$income[!lfs_samp_ind$rep_ind] <- 0

```

```

qvar8 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind),
  sampling_stages = 3,
  id = c("id_area", "id_dwel", "id_ind"),
  parent_id = c("id_area", "id_dwel"),
  dissemination_dummy = "rep_ind",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel", "w_ind"),
  # NR correction at stages 2 and 3
  nrc_weight = list(NULL, "w_nrc", "w_nrc"),
  response_dummy = list(NULL, "rep", "rep_ind"),
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - 3-stage sampling design
#> - stage 1: simple random sampling WITHOUT stratification
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> - stage 3: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: At stage 3, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_dwel).
#> Warning: The following strata at stage 3 contain less than two sampled units:
  ↳ D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1,
  ↳ D02653:1, D03338:1 and 18 more. They are excluded from the variance estimation
  ↳ process (but kept for point estimates).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

```

Note that the *same variable name* may be used at different stages: each element of the list is looked up in the data.frame of its own stage (`w_nrc` in `lfs_samp_dwel` at stage 2, `w_nrc` in `lfs_samp_ind` at stage 3).

```

var_srs_three <- function(y, smp_area, smp_dwel, smp_ind) {
  variance <- list()

  # Stage 3 (individuals): NR then (exhaustive) sampling
  resp_ind <- smp_ind %>% filter(rep_ind == 1)

  variance[["nr-ind"]] <- var_pois(
    y[resp_ind$id_ind, , drop = FALSE],
    pik = resp_ind$prep_ind,
    # Rao weight: cumulated design weight down to stage 3
    w = resp_ind$w_nrc_dwel * resp_ind$w_area * resp_ind$w_ind
  )

  y <- add_zero(y, rownames = smp_ind$id_ind)
  y_dwel <- sum_by(y / smp_ind$prep_ind, by = smp_ind$id_dwel, w = smp_ind$w_ind)

  # Stage 2 (dwellings): NR then sampling
  resp_dwel <- smp_dwel %>% filter(rep == 1)

```

```

variance[["nr"]] <- var_pois(
  y_dwel[resp_dwel$id_dwel, , drop = FALSE],
  pik = resp_dwel$prep,
  w = resp_dwel$w_dwel * resp_dwel$w_area
)

y_dwel <- add_zero(y_dwel, rownames = smp_dwel$id_dwel)

variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
  sub <- smp_dwel %>% filter(id_area == .x)
  w_area <- unique(sub$w_area)
  var_srs(
    y_dwel[sub$id_dwel, , drop = FALSE] / sub$prep,
    pik = sub$pik_dwel
  ) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y_dwel / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

# Stage 1 (areas)
variance[["srs-area"]] <- var_srs(
  y_area[smp_area$id_area, , drop = FALSE],
  pik = 1 / smp_area$w_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar8b <- define_variance_wrapper(
  variance_function = var_srs_three,
  technical_data = list(smp_area = lfs_samp_area,
    smp_dwel = lfs_samp_dwel,
    smp_ind = lfs_samp_ind),
  reference_id = lfs_samp_ind$id_ind,
  reference_weight = lfs_samp_ind$w_final,
  default_id = "id_ind"
)

qvar8(lfs_samp_ind, total(income))
#> Warning: 15 observations do not match any responding units of the survey. They are
#> discarded.
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 53 194416.5 793896723 28176.17 14.49268 139192.3 249640.8
qvar8b(lfs_samp_ind, total(income))
#>      nr.ind      nr srs.dwel srs.area
#> 1 977719 4066099 31686319 757166586
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 194416.5 793896723 28176.17 14.49268 139192.3 249640.8

```

The Rao weight of the individual-stage non-response term deserves a pause: it is the cumulated weight of everything above and including the individual design stage — $w_area * w_nrc_dwel * w_ind$ — where the dwelling contribution is the *corrected* (post-NR) conditional weight. This is exactly what `qvar_ms()` builds

internally as `upper_weight * sampling_weight`.

2.10 Case 9: calibration at an intermediate stage

Calibration does not have to occur at the final stage. With `calibration_stage = 2`, the calibration residuals are computed at the dwelling stage, on the estimated dwelling totals aggregated from the individual stage; the stages below the calibration stage are left unresidualized. This is the first-order (Deville) linearization of the calibrated estimator: the residual technique applies at the stage of the units whose weights were calibrated.

```
qvar9 <- qvar_ms(
  data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind),
  sampling_stages = 3,
  id = c("id_area", "id_dwel", "id_ind"),
  parent_id = c("id_area", "id_dwel"),
  dissemination_dummy = "rep_ind",
  dissemination_weight = "w_final",
  sampling_weight = c("w_area", "w_dwel", "w_ind"),
  nrc_weight = list(NULL, "w_nrc", "w_nrc"),
  response_dummy = list(NULL, "rep", "rep_ind"),
  # Calibration at stage 2 (dwellings), CUMULATED weight at that stage
  calibration_stage = 2,
  calibration_weight = "w_final",
  calibration_var = "strata_dwel",
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - 3-stage sampling design
#> - stage 1: simple random sampling WITHOUT stratification
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> - stage 3: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#> - calibration on margins at stage 2
#> Note: The following calibration variables are qualitative (factor, character):
  ↳ strata_dwel. They will be automatically discretized.
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: At stage 3, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_dwel).
#> Warning: The following strata at stage 3 contain less than two sampled units:
  ↳ D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1,
  ↳ D02653:1, D03338:1 and 18 more. They are excluded from the variance estimation
  ↳ process (but kept for point estimates).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

```
var_srs_three <- function(y, smp_area, smp_dwel, smp_ind) {

  variance <- list()

  # Stage 3 (individuals): NR, no residuals here
```

```

resp_ind <- smp_ind %>% filter(rep_ind == 1)

variance[["nr-ind"]] <- var_pois(
  y[resp_ind$id_ind, , drop = FALSE],
  pik = resp_ind$prep_ind,
  w = resp_ind$w_nrc_dwel * resp_ind$w_area * resp_ind$w_ind
)

y <- add_zero(y, rownames = smp_ind$id_ind)
y_dwel <- sum_by(y / smp_ind$prep_ind, by = smp_ind$id_dwel, w = smp_ind$w_ind)

# Stage 2 (dwellings): calibration residuals on the estimated totals,
# then NR and sampling
resp_dwel <- smp_dwel %>% filter(rep == 1)

x_cal <- model.matrix(~ strata_dwel, data = resp_dwel)
y_dwel[resp_dwel$id_dwel, ] <- res_cal(
  y_dwel[resp_dwel$id_dwel, ],
  x = x_cal,
  w = resp_dwel$w_final
)

variance[["nr"]] <- var_pois(
  y_dwel[resp_dwel$id_dwel, , drop = FALSE],
  pik = resp_dwel$prep,
  w = resp_dwel$w_dwel * resp_dwel$w_area
)

y_dwel <- add_zero(y_dwel, rownames = smp_dwel$id_dwel)

variance[["srs-dwel"]] <- unique(smp_dwel$id_area) %>% map(~{
  sub <- smp_dwel %>% filter(id_area == .x)
  w_area <- unique(sub$w_area)
  var_srs(
    y_dwel[sub$id_dwel, , drop = FALSE] / sub$prep,
    pik = sub$pik_dwel
  ) * w_area
}) %>% unlist %>% sum

y_area <- sum_by(y_dwel / smp_dwel$prep, by = smp_dwel$id_area, w = smp_dwel$w_dwel)

variance[["srs-area"]] <- var_srs(
  y_area[smp_area$id_area, , drop = FALSE],
  pik = 1 / smp_area$w_area
)

print(as.data.frame(variance))
Reduce(`+`, variance)
}

qvar9b <- define_variance_wrapper(
  variance_function = var_srs_three,
  technical_data = list(smp_area = lfs_samp_area,
    smp_dwel = lfs_samp_dwel,

```

```

        smp_ind = lfs_samp_ind),
reference_id    = lfs_samp_ind$id_ind,
reference_weight = lfs_samp_ind$w_final,
default_id     = "id_ind"
)

qvar9(lfs_samp_ind, total(income))
#> Warning: 15 observations do not match any responding units of the survey. They are
→ discarded.
#>      call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 53 194416.5 858017146 29291.93 15.06658 137005.4 251827.7
qvar9b(lfs_samp_ind, total(income))
#>      nr.ind      nr srs.dwel srs.area
#> 1 977719 1380808 16840377 838818242
#>      call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 194416.5 858017146 29291.93 15.06658 137005.4 251827.7

```

Note that `calibration_weight` refers here to a variable of the *stage 2* file (`lfs_samp_dwel$w_final`): the cumulated weight at the calibration stage, i.e. `w_area * w_nrc` at the dwelling stage.

3 Estimation diagnostics

Beyond point estimates and confidence intervals, a variance estimation run carries a wealth of methodological information: where the variance comes from, what the design costs in precision, whether the confidence intervals can be trusted, whether the non-response correction is under control. `qvar_ms()` exposes this information through the `diagnostics` technical parameter of the variance wrapper.

Because `diagnostics` is a *technical parameter* (see `define_variance_wrapper()`), it is an argument of the wrapper itself: it can be switched on for a single estimation, without redefining anything and at negligible cost, since all precalculated data are shared.

```

# Nothing by default (batch-friendly)
precision(lfs_samp_dwel, total(income))

# Everything
precision(lfs_samp_dwel, total(income), diagnostics = "all") # or TRUE

# A subset, with partial matching
precision(lfs_samp_dwel, total(income), diagnostics = c("comp", "deff"))

```

The five available diagnostics are "components", "deff", "dof", "nr" and "domains". Two practical remarks before going through them. First, diagnostics are printed to the console (standard output) to be used during interactive work. Second, when several statistics are requested in one call, diagnostics are reported per statistic, numbered in the order of the call.

The wrappers below are those defined in cases 6 and 9 of this vignette.

3.1 Variance components ("components")

The Rao (1975) decomposition implemented by `qvar_ms()` is additive: the total variance is the sum of one sampling component per stage (`sampling_stage1`, `sampling_stage2`, ...) plus one non-response component

per corrected stage (`nr_stage2`, ...), plus, when the Poisson treatment of single-unit strata is enabled, the corresponding `sampling_pois_stage*` terms. The "components" diagnostic prints this decomposition with the share of each component in the total:

```
qvar6(lfs_samp_dwel, total(income), diagnostics = "components")
#> Warning: 32 observations do not match any responding units of the survey. They are
→ discarded.
#>
#> -- Variance components: calibration effect --
#>
#>          with calib.      share          without      effect
#> nr_stage2      6.592e+05    0.3 %      3.213e+06    -79.5 %
#> sampling_stage2 8.082e+06    3.3 %      2.2e+07     -63.3 %
#> sampling_stage1 2.396e+08   96.5 % #####      1.029e+08   +132.8 %
#> TOTAL          2.483e+08  100.0 % #####      1.281e+08    +93.8 %
#>
#> Calibration multiplies the variance by 1.938 (deft x 1.392).
#>
#>          call n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 248325893 15758.36 8.274313 159563.3 221335
```

When the wrapper includes a calibration step, the diagnostic automatically computes the variance of the *uncalibrated* estimator alongside (the Horvitz-Thompson estimator corrected for non-response), by carrying a duplicated, non-residualized copy of the linearized variable through the same single pass. The “effect” column is thus the per-component effect of calibration. Note: the “without” column is the variance of a *different estimator* (with a different point estimate since it involves no calibration step).

Warning signs.

- *A non-zero calibration effect below the calibration stage.* Components of stages strictly below the calibration stage are computed before the residualization in the upward loop: they must display exactly +0.0 %. Anything else indicates a bug or a misdeclared `calibration_stage`.
- *A dominant first-stage component.* When `sampling_stage1` accounts for most of the variance, the clustering of the design is the binding constraint: increasing the number of final units per PSU will barely improve precision (see the `roh` diagnostic below), whereas increasing the number of PSUs will.
- *A positive calibration effect on a variable supposedly related to the margins.* The variance reduction of calibration is only guaranteed for variables that the calibration variables explain. A + effect on a target variable of the survey suggests reviewing the choice of margins.

3.2 Design effect ("deff")

The design effect (Kish, 1965) is the ratio of the variance under the actual design to the variance of the same estimator under a simple random sample of the same size:

$$\text{deff}(\hat{\theta}) = \frac{\hat{V}_{\text{design}}(\hat{\theta})}{\hat{V}_{\text{SRS}}(\hat{\theta})}, \quad \text{deft} = \sqrt{\text{deff}}, \quad n_{\text{eff}} = n/\text{deff}.$$

The denominator is computed by `var_srs()` with a constant inclusion probability $n/\hat{N}$, applied to the *same linearized variable* as the numerator — so that the design effect is automatically valid for means, ratios and domain estimates, with the following conventions: n is the number of disseminated units and $\hat{N}$ the sum of the full design weights.

The diagnostic also prints the classical Kish decomposition: the unequal weighting effect $\text{deff}_w = 1 + \text{CV}^2(w)$ (Kish, 1992), computed exactly from the weights, and the residual clustering effect $\text{deff}_c = \text{deff}/\text{deff}_w$, from which the rate of homogeneity is derived:

$$\text{roh} = \frac{\text{deff}_c - 1}{\bar{m} - 1},$$

with $\bar{m}$ the mean number of final units per primary sampling unit. `roh` estimates the intra-cluster correlation of the study variable and can be used as an input for designing future waves of the survey.

```
qvar6(lfs_samp_dwel, total(income), diagnostics = "deff")
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   ↪   discarded.
#>
#> -- Design effect (Kish) --
#> Variance under the actual design, relative to a simple random sample
#> of the same size (n = 48 disseminated units, estimated population N = 9 619).
#>
#>   statistic 1:  deff = 1.163   |  deff = 1.078   |  effective n = 41
#>
#> A deff above 1 quantifies the precision cost of the design (clustering,
#> unequal weights); below 1, stratification and calibration gains dominate.
#>
#>   Kish decomposition:  deff_weights = 1.000 (CV of weights = 0.00)
#>   statistic 1:  deff_cluster = 1.163   |  roh = 0.0109 (mean cluster size = 16.0)
#>
#>           call  n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 248325893 15758.36 8.274313 159563.3 221335
```

Warning signs.

- *deff far from 1 on a design that is (nearly) a SRS.* This is the acid test of the whole computation: on case 1 of this vignette, the diagnostic must print `deff = 1.000` up to rounding. A systematic departure signals a weight convention problem, not a design feature.
- *A large weighting effect.* $\text{CV}(w)$ above roughly 1 (i.e. $\text{deff}_w > 2$) usually reveals extreme weights — often produced by the non-response correction (see the `"nr"` diagnostic) — and calls for trimming or coarser reweighting cells.
- *A large roh with a large mean cluster size.* Since $\text{deff}_c \approx 1 + (\bar{m} - 1) \text{roh}$, precision is bought by more PSUs, not by more units per PSU. Slightly *negative* values of `roh` are possible and benign: they reflect stratification gains that the Kish decomposition attributes to the clustering term.
- *Caveat.* The weights used in the decomposition are reconstructed from the design and non-response information and ignore the calibration g-weights; on strongly calibrated surveys, deff_w is a slight approximation.

3.3 Degrees of freedom ("dof")

Analytical variance estimators for clustered designs behave, to first order, like variance estimators with

$$\text{df} = (\text{number of PSUs}) - (\text{number of first-stage strata})$$

degrees of freedom. The normal-based confidence intervals produced by `gustave` implicitly assume these degrees of freedom to be large. The `"dof"` diagnostic prints the rule-of-thumb `df` and, when it is small, the width correction a Student-based interval would apply:

```
-- Degrees of freedom --
df = 80 PSUs - 2 strata = 78
Large-sample normal intervals are adequate.
```


versus, in our sample, we only have 4 PSU and no first-stage strata:

```
qvar6(lfs_samp_dwel, total(income), diagnostics = "dof")
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   ↪   discarded.
#>
#> -- Degrees of freedom --
#>   df = 3 PSUs - 1 strata = 2
#>   Normal-based CIs are optimistic: a t-based CI would be 119.5 % wider.
#>
#>           call n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 248325893 15758.36 8.274313 159563.3 221335
```

Warning signs.

- *df below ~30.* Normal intervals undercover; report t-based intervals or at least mention the issue. Below ~10, the variance estimate itself is unstable.
- *Domain estimation concentrated in few PSUs.* The nominal df is an upper bound: for a domain present in only a handful of PSUs, the effective df is closer to the number of PSUs containing the domain. Cross-read with the "domains" diagnostic.

3.4 Non-response diagnostics ("nr")

For each stage with a non-response correction, the diagnostic prints the unweighted and weighted response rates, the distribution of the estimated response probabilities, and the Kish inflation factor $1 + CV^2(1/\hat{p})$ applied to the variance by the dispersion of the non-response weights:

```
qvar9(lfs_samp_ind, total(income), diagnostics = "nr")
#> Warning: 15 observations do not match any responding units of the survey. They are
#>   ↪   discarded.
#>
#> -- Non-response, stage 2 --
#>   response rate: 60.0 % (weighted 60.0 %)
#>   response prob.: min 0.600 | median 0.600 | max 0.600
#>   reweighting inflation (Kish): x 1.000 on the NR-phase variance
#>
#> -- Non-response, stage 3 --
#>   response rate: 77.9 % (weighted 77.9 %)
#>   response prob.: min 0.800 | median 0.800 | max 0.800
#>   reweighting inflation (Kish): x 1.000 on the NR-phase variance
#>
#>           call n      est variance      std      cv      lower upper
#> 1 total(y = income) 53 194416.5 858017146 29291.93 15.06658 137005.4 251827.7
```

Warning signs.

- *Very small minimum response probabilities.* A $\hat{p}_{\min}$ below roughly 0.1 means some units carry a tenfold non-response weight: a single such unit can dominate a domain estimate.
- *A gap between unweighted and weighted response rates.* It indicates that response propensity is correlated with the design weights — the non-response mechanism differs across sampling strata or stages, which is worth modelling explicitly (Särndal and Lundström, 2005).
- *An inflation factor well above 1.* The variance cost of reweighting itself; combined with a weak relation between the reweighting cells and the study variable, it is a net loss. (Little and Vartivarian, 2005)
- At the last stage, the unweighted rate includes out-of-scope units in its denominator when a `scope_dummy` is declared; interpret accordingly.

3.5 Domain sample sizes ("domains")

Domain estimation through `by` and `where` is deceptively easy: the linearized variable is simply zeroed outside the domain, and the machinery runs unchanged. The "domains" diagnostic counts the non-zero observations behind each requested statistic and flags small domains when observations falls below 50:

```
qvar6(lfs_samp_dwel,
  "Income, stratum S1" = total(income, where = strata_dwel == "S1"),
  "Income, stratum S2" = total(income, where = strata_dwel == "S2"),
  diagnostics = "domains"
)
#> Warning: 32 observations do not match any responding units of the survey. They are
  ↳ discarded.
#>
#> -- Domain sample sizes --
#>   statistic 1: 15 non-zero observations ** small domain: estimate may be unstable **
#>   statistic 2: 13 non-zero observations ** small domain: estimate may be unstable **
#>           label                                call  n      est
#> 1 Income, stratum S1 total(y = income, where = strata_dwel == "S1") 15 54622.71
#> 2 Income, stratum S2 total(y = income, where = strata_dwel == "S2") 13 62277.33
#>   variance      std      cv   lower   upper
#> 1 100587981 10029.356 18.361148 34965.53 74279.88
#> 2  6409195  2531.639  4.065105 57315.41 67239.25
```

Warning signs.

- *Domains below 50 observations.* Both the estimate and its estimated variance are unstable; this is the natural boundary where design-based estimation gives way to small area methods.
- *Small domains combined with a low df.* The two previous caveats compound: a domain concentrated in a handful of PSUs has both an unstable variance and fewer effective degrees of freedom than the nominal rule suggests.

3.6 A reading grid

Symptom	Likely cause	First action
deff far from 1 under a (near-)SRS design dominant <code>sampling_stage1</code> component	weight convention error (conditional vs cumulated) clustering cost	check " <code>nr</code> " probabilities and the calibration g-ratio warning more PSUs, not more units per PSU; read <code>roh</code>
calibration effect expected +0.0 % but non-zero below the calibration stage	misdeclared <code>calibration_stage</code> or bug	check the wrapper with <code>inspect_wrapper()</code>
large <code>deff_weights</code>	extreme weights	inspect " <code>nr</code> " minimum probabilities; trim or collapse cells
df < 30	few PSUs	report t-based intervals; consider replication cross-check
small-domain flag	thin domain	precision thresholds; small area estimation

4 Inspecting a variance wrapper

A variance wrapper is a peculiar R object: a function that carries, frozen in its environment and attributes, the complete methodological description of a survey — identifiers, weights, strata, response probabilities, calibration data, precalculated matrices. This design is what makes wrappers so convenient to *use* (one self-contained object, no need to re-specify the design at each estimation), but it also makes them opaque: printing the function body tells you nothing about the survey it encodes.

`qvar_notes()` recalls the definition, the notes and the warnings emitted when the wrapper was defined. This is often enough to answer the most common question about a wrapper someone else defined: “*which features does it take into account, and did anything unusual happen at definition time?*” — the welcome message, the automatic coercions, the strata crossing notes and the single-unit strata warnings are all there.

```
qvar_notes(qvar9)
#> Wrapper defined by:
#>
#> qvar_ms(data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind),
#>   sampling_stages = 3, id = c("id_area", "id_dwel", "id_ind"),
#>   parent_id = c("id_area", "id_dwel"), dissemination_dummy = "rep_ind",
#>   dissemination_weight = "w_final", sampling_weight = c("w_area",
#>     "w_dwel", "w_ind"), nrc_weight = list(NULL, "w_nrc",
#>     "w_nrc"), response_dummy = list(NULL, "rep", "rep_ind"),
#>   calibration_stage = 2, calibration_weight = "w_final", calibration_var =
  ↪ "strata_dwel",
#>   define = TRUE)
#>
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 3-stage sampling design
#>   - stage 1: simple random sampling WITHOUT stratification
#>   - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↪ stage above), with non-response correction through reweighting
#>   - stage 3: simple random sampling WITHOUT stratification (within the units of the
  ↪ stage above), with non-response correction through reweighting
#>   - calibration on margins at stage 2
#> Note: The following calibration variables are qualitative (factor, character):
  ↪ strata_dwel. They will be automatically discretized.
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↪ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: At stage 3, units are drawn within the units of the stage above: the effective
  ↪ strata used for variance estimation are the crossing of the parent unit (id_dwel).
#> Warning: The following strata at stage 3 contain less than two sampled units:
  ↪ D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1,
  ↪ D02653:1, D03338:1 and 18 more. They are excluded from the variance estimation
  ↪ process (but kept for point estimates).
```

`inspect_wrapper()` opens the box. Given a wrapper produced by `qvar_ms()`, it prints a structured console summary and generates an HTML report describing the reference population, each sampling stage, the methodological notes, and — most importantly — the exact variance estimation pipeline that will be executed. It serves three audiences at once:

- **documentation** — the report is a self-contained methodological description of the survey, generated from the object itself and therefore always in sync with what is actually computed; it can be attached to quality reports or handed to a new colleague;

- **audit** — a reviewer can check, without reading any code, that the declared design (stages, strata, non-response corrections, calibration stage) matches the survey documentation;
- **debugging** — most of the estimation problems discussed in this vignette (weight conventions, excluded strata, response probability ranges) are visible at a glance in the per-stage cards, *before* any estimation is run.

```
inspect_wrapper(qvar9, plot = FALSE)
#> Warning in inspect_wrapper(qvar9, plot = FALSE): This wrapper does look like a
#>   ↳   quars() wrapper: technical_data
#>   contain a 'stages' list and a 'calib' element.
#>
#> =====
#>   MULTISTAGE VARIANCE WRAPPER INSPECTION (quars)
#> =====
#>
#> -- Reference population --
#>   Analysis id:      id_ind
#>   N units:         53
#>   First 5 ids:      I00128, I00416, I01216, I01514, I02585
#>   Positive weight range: [248.9, 251]
#>   Zero-weight:      0 units
#>
#> -- Sampling design: 3 stage(s) --
#>
#>   Stage 1 (primary units)
#>     Units:          4
#>     Strata:          1
#>     Cond. sampling weight: [19.92, 19.92]
#>     Upper weight:      1 (top stage)
#>     Non-response corr.: no
#>     Calibration:      no
#>
#>   Stage 2
#>     Units:          80
#>     Strata:          4 (parent x declared strata)
#>     Cond. sampling weight: [6, 6.05]
#>     Upper weight:      [19.92, 19.92]
#>     Non-response corr.: YES - 48 respondents, response prob. in [0.6, 0.6] (mean
#>   ↳   0.6)
#>     Calibration:      YES - 48 units, 3 variables
#>     Aggregates to:     4 parent units (stage 1)
#>
#>   Stage 3 (final units)
#>     Units:          68 (28 excluded: single-unit strata)
#>     Strata:          0 (parent x declared strata)
#>     Exhaustive units: 40 (pik = 1, discarded by varDT)
#>     Cond. sampling weight: [1, 1]
#>     Upper weight:      [199.2, 200.8]
#>     Non-response corr.: YES - 53 respondents, response prob. in [0.8, 0.8] (mean
#>   ↳   0.8)
#>     Calibration:      no
#>     Aggregates to:     48 parent units (stage 2)
#>
#> -- Technical parameters --
#>   single_unit_strata <character> length 1
```

```

#> diagnostics: NULL
#>
#> -- Variance estimation pipeline (execution order) --
#> 1 L3 [y] add_zero() Expand y to all stage-3 sampled units (zeros elsewhere)
#>      68 units
#> 2 L3 [V_NR] var_pois() Non-response variance (Poisson) - stage 3
#>      53 responding units, response prob. in [0.8, 0.8] - weight w = upper_weight
#>      × conditional sampling weight
#> 3 L3 [y] Diagonal() Non-response reweighting - stage 3
#>      expansion of the 53 responding units
#> 4 L3 [V_SMP] var_srs() Sampling variance (stratified SRS) - stage 3
#>      40 units (28 excluded: single-unit strata), 0 strata (parent × declared
#>      strata) - w = upper_weight
#> 5 L3 [AGG] sum_by() Aggregate to stage 2 (estimated totals per parent unit)
#>      48 parent units
#> 6 L2 [y] add_zero() Expand y to all stage-2 sampled units (zeros elsewhere)
#>      80 units
#> 7 L2 [CAL] res_cal() Calibration residuals
#>      48 calibrated units, 3 calibration variables (discretized)
#> 8 L2 [V_NR] var_pois() Non-response variance (Poisson) - stage 2
#>      48 responding units, response prob. in [0.6, 0.6] - weight w = upper_weight
#>      × conditional sampling weight
#> 9 L2 [y] Diagonal() Non-response reweighting - stage 2
#>      expansion of the 48 responding units
#> 10 L2 [V_SMP] var_srs() Sampling variance (stratified SRS) - stage 2
#>      80 units, 4 strata (parent × declared strata) - w = upper_weight
#> 11 L2 [AGG] sum_by() Aggregate to stage 1 (estimated totals per parent unit)
#>      4 parent units
#> 12 L1 [y] add_zero() Expand y to all stage-1 sampled units (zeros elsewhere)
#>      4 units
#> 13 L1 [V_SMP] var_srs() Sampling variance (stratified SRS) - stage 1
#>      4 units, 1 strata - w = 1 (top stage)
#> 14 [SUM] Reduce() Sum variance components
#>      5 components: nr_stage3, sampling_stage3, nr_stage2, sampling_stage2,
#>      sampling_stage1

```

The function prints a console summary and opens an HTML report in the browser (use `plot = FALSE` for console-only, or `file =` to choose where the report is written, e.g. to archive it alongside the survey documentation).

The HTML report presents the same information in a series of cards, described below with their intended screenshots.

4.1 Reference population

The first card describes the **reference population**: the set of units, frozen at definition time, on which every estimation will be based — for a `qvar_ms()` wrapper, the disseminated units of the last sampling stage, together with their dissemination weights. It is best read as the *contract* between the wrapper and any estimation file: whatever `data.frame` is passed at estimation time is matched against these identifiers (extra rows are discarded, missing units count as zeros, and a differently ordered file is silently reordered — the warnings discussed earlier in this vignette all originate from this matching step). The card reports the analysis identifier (the column the estimation file must contain), the number of reference units, a preview of the first identifiers and weights, the range of the positive weights and the number of zero-weight units.

Reference Population	
Analysis id	id_ind
Sampling stages	3
Number of units	53
First 5 ids	I00128, I00416, I01216, I01514, I02585
Positive weight range	[249.3, 8158]
Zero-weight units	0

Figure 1: Reference population card of the HTML report.

4.2 Per-stage cards

Stage 1 Primary units	Stage 2 Intermediate units	Stage 3 Final units
Units 4	Units 80	Units 68 (28 excluded: single-unit strata)
Strata 1	Strata 4 (parent × declared strata)	Strata 0 (parent × declared strata)
Conditional sampling weight [19.92, 19.92]	Conditional sampling weight [6, 6.05]	Exhaustive units (pik = 1) 40
Upper weight 1 (top stage)	Upper weight [19.92, 19.92]	Conditional sampling weight [1, 1]
Non-response correction no	Non-response correction YES 48 respondents · response prob. in [0.6, 0.6] (mean 0.6)	Upper weight [199.2, 200.8]
Calibration no	Calibration YES 48 units · 2 calibration variables	Non-response correction YES 53 respondents · response prob. in [0.02461, 0.7989] (mean 0.359)
	Aggregates to 4 parent units (stage 1)	Calibration no
		Aggregates to 48 parent units (stage 2)

Figure 2: Per-stage cards of the HTML report.

Each stage card summarizes what `qvar_ms()` retained from the declaration: number of sampled units and excluded units (single-unit strata), number of effective strata (with the parent-crossing note from stage 2 onwards), range of the conditional sampling weights and of the cumulated Rao weight, non-response correction (with the range of the estimated response probabilities) and calibration. These are computed from the *actual* technical data of the wrapper, not from the arguments of the call — which makes them a plausibility check as much as a description. Two examples of what to look for, both drawn from real debugging sessions:

- **response probabilities outside a plausible range** — a response prob. in [0.07, 0.12] where the survey documentation says the response rate is around 70 % is the signature of a *cumulated* non-response weight declared where a *conditional* one is expected (see the Conventions section); the estimation would run and silently inflate every variance component;
- **an Upper (Rao) weight different from 1 at stage 1**, or a conditional weight range at odds with the sampling fractions of the design, points to the same family of weight convention errors.

Methodological notes

```
CALL qvar_ms(data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind), sampling_stages = 3,
  id = c("id_area", "id_dwel", "id_ind"), parent_id = c("id_area", "id_dwel"),
  dissemination_dummy = "rep_ind", dissemination_weight = "w_final", sampling_weight = c("w_area",
    "w_dwel", "w_ind"), nrc_weight = list(NULL, "w_nrc", "w_nrc"), response_dummy = list(NULL,
    "rep", "rep_ind"), calibration_stage = 2, calibration_weight = "w_final",
  calibration_var = "strata_dwel", define = TRUE)
```

INFO Survey variance estimation with the gustave package

The following features are taken into account:

- 3-stage sampling design
- stage 1: simple random sampling WITHOUT stratification
- stage 2: simple random sampling WITHOUT stratification (within the units of the stage above), with non-response correction through reweighting
- stage 3: simple random sampling WITHOUT stratification (within the units of the stage above), with non-response correction through reweighting
- calibration on margins at stage 2

NOTE The following calibration variables are qualitative (factor, character): strata_dwel. They will be automatically discretized.

NOTE At stage 2, units are drawn within the units of the stage above: the effective strata used for variance estimation are the crossing of the parent unit (id_area).

NOTE At stage 3, units are drawn within the units of the stage above: the effective strata used for variance estimation are the crossing of the parent unit (id_dwel).

WARNING The following strata at stage 3 contain less than two sampled units: D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1, D02653:1, D03338:1 and 18 more. They are excluded from the variance estimation process (but kept for point estimates).

Figure 3: Methodological notes card: notes and warnings emitted at definition time.

4.3 Methodological notes

The Methodological notes card replays, colour-coded, everything that was printed when the wrapper was defined: the definition call itself, the welcome message (INFO), the automatic coercions and strata-crossing notes (NOTE, blue), and the warnings (orange) — excluded single-unit strata, unequal weights replaced by stratum means, suspicious calibration g-ratios.

4.4 Pipeline table and diagram

Pipeline Steps (execution order: last stage → first stage)						
#	Stage	Type	Function	Description	Component	Formula
1	L3	transition	add_zero()	Expand y to all stage-3 sampled units (zeros elsewhere)		
2	L3	variance_nr	var_pois()	Non-response variance (Poisson) — stage 3	nr_stage3	$\sum_{i \in r_3} w_{hi}^{nr} s_{hi}^2 (1 - \hat{p}_i) \left(\frac{p_i}{\hat{p}_i} \right)^2$
3	L3	transition	Diagonal()	Non-response reweighting — stage 3		$y_i \leftarrow y_i / \hat{p}_i$
4	L3	variance_sampling	var_srs()	Sampling variance (stratified SRS) — stage 3	sampling_stage3	$\sum_h w_h^{nr} N_h^2 \left(1 - \frac{n_h}{N_h} \right) \frac{s_h^2}{n_h}$
5	L3	aggregation	sum_by()	Aggregate to stage 2 (estimated totals per parent unit)		$\hat{Y}_j = \sum_{i \in j} \frac{y_i}{w_{hi}}$
6	L2	transition	add_zero()	Expand y to all stage-2 sampled units (zeros elsewhere)		
7	L2	calibration	res_cal()	Calibration residuals		$y_i - x_i' \hat{\beta}$
8	L2	variance_nr	var_pois()	Non-response variance (Poisson) — stage 2	nr_stage2	$\sum_{i \in r_2} w_{hi}^{nr} s_{hi}^2 (1 - \hat{p}_i) \left(\frac{p_i}{\hat{p}_i} \right)^2$
9	L2	transition	Diagonal()	Non-response reweighting — stage 2		$y_i \leftarrow y_i / \hat{p}_i$
10	L2	variance_sampling	var_srs()	Sampling variance (stratified SRS) — stage 2	sampling_stage2	$\sum_h w_h^{nr} N_h^2 \left(1 - \frac{n_h}{N_h} \right) \frac{s_h^2}{n_h}$
11	L2	aggregation	sum_by()	Aggregate to stage 1 (estimated totals per parent unit)		$\hat{Y}_j = \sum_{i \in j} \frac{y_i}{w_{hi}}$
12	L1	transition	add_zero()	Expand y to all stage-1 sampled units (zeros elsewhere)		
13	L1	variance_sampling	var_srs()	Sampling variance (stratified SRS) — stage 1	sampling_stage1	$\sum_h w_h^{nr} N_h^2 \left(1 - \frac{n_h}{N_h} \right) \frac{s_h^2}{n_h}$
14		summation	Reduce()	Sum variance components		$\hat{V} = \hat{V}_{nr_stage3} + \hat{V}_{sampling_stage3} + \hat{V}_{nr_stage2} + \hat{V}_{sampling_stage2} + \hat{V}_{nr_stage1} + \hat{V}_{sampling_stage1}$

Figure 4: The variance estimation pipeline

This is the heart of the report. One essential design decision deserves a word: the pipeline is derived from

the **technical data** of the wrapper, not from the code of the variance function. The code is a static loop with conditional blocks — `res_cal` and `var_pois` appear in it whether or not they will run, and `var_srs` appears once although it runs once *per stage*. What determines the actual execution is the content of **technical_data**: the inspector therefore rebuilds the pipeline from it, so that the diagram shows exactly the steps that will execute, stage by stage — a calibration node only in the calibration stage’s subgraph, a non-response node only at corrected stages, one sampling variance node per stage.

Each variance node carries the name of its component (`nr_stage2`, `sampling_stage1`, ...). These names are the same as those printed by the `diagnostics = "components"` output and stored in the variance list: the diagram is thus a legend for the numerical decomposition — when a component dominates, the diagram shows where in the design it comes from.

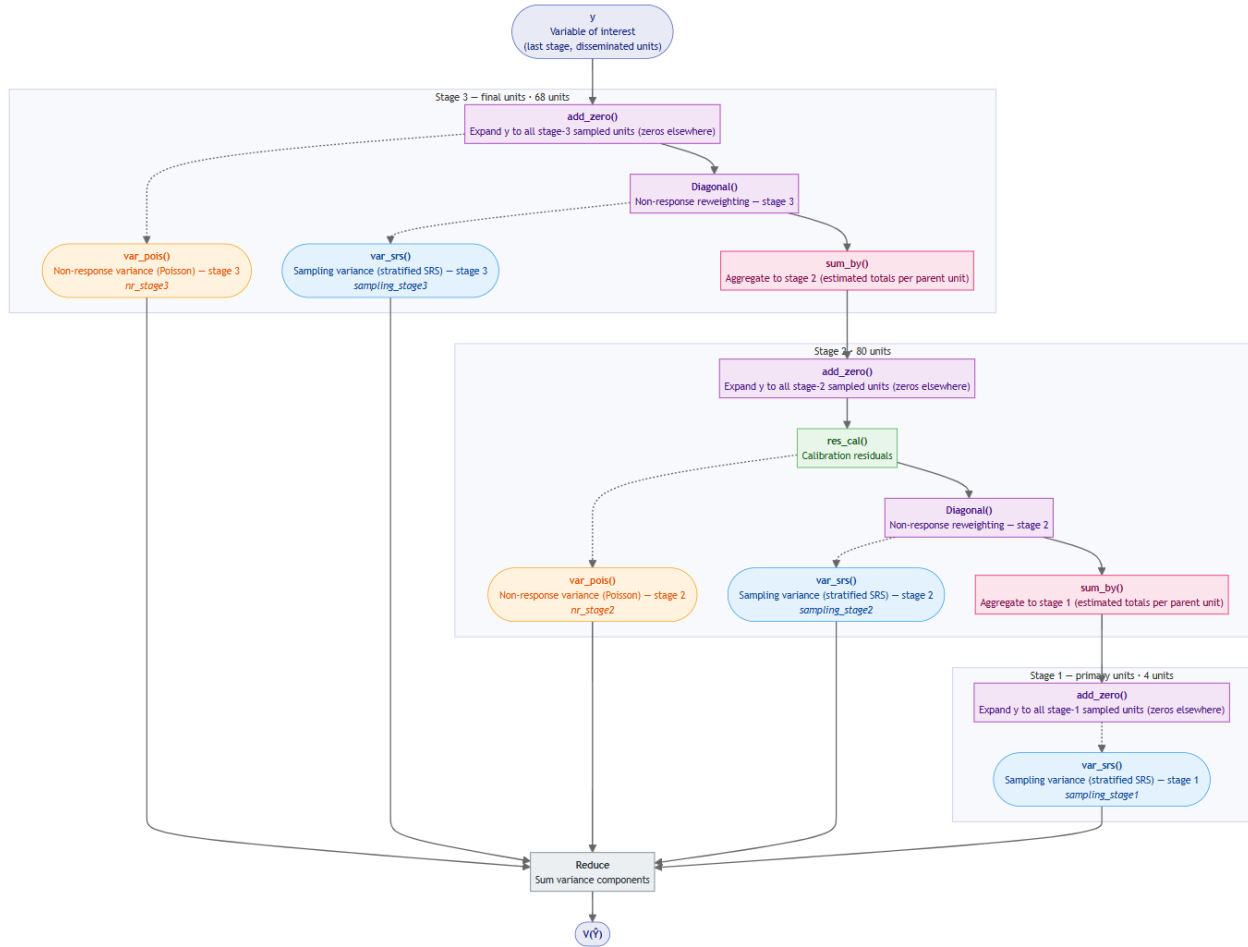


Figure 5: The variance estimation diagram

4.5 Technical parameters and raw technical data

The technical parameters cards lists the switchable arguments of the wrapper (`single_unit_strata`, `diagnostics`) with their defaults — the knobs a user can turn at estimation time without redefining anything.

The last card is a collapsible tree of the raw technical data, descending recursively into the per-stage lists (identifiers, weights, exclusion flags, precalculated matrices) with types, dimensions and value previews. It is the map for advanced users who want to go beyond the report.

Technical Parameters			
Name	Type	Dimension	Preview
single_unit_strata	character	length 1	excl ^u de
diagnostics	NULL	-	Not provided

Figure 6: Technical parameters.

▼ Raw technical data structure (click to expand)			
Name	Type	Dimension	Preview
technical_data	list	2 elements	
L stages	list	3 elements	
L [1[]]	list	5 elements	
L samp	list	3 elements	
L id	character	length 4	A005=A005, A007=A007, A016=A016
L exclude	logical	length 4	A005=FALSE, A007=FALSE, A016=FALSE
L precalc	list	7 elements	
L nrc	NULL	-	
L upper_weight	numeric	length 4	A005=1, A007=1, A016=1
L agg_weight	numeric	length 4	A005=19.92, A007=19.92, A016=19.92
L parent	NULL	-	
L [2[]]	list	5 elements	
L samp	list	3 elements	
L id	character	length 80	D00012=D00012, D00131=D00131, D00165=D00165
L exclude	logical	length 80	D00012=FALSE, D00131=FALSE, D00165=FALSE
L precalc	list	7 elements	

Figure 7: Raw technical data.

Combined with `qvar_notes()` for the methodological notes and the **diagnostics** technical parameter for estimation-time output, the inspector completes a three-level tooling: the *log* tells what happened at definition, the *report* describes what the wrapper is, and the *diagnostics* tell what each estimation does.

5 Conclusion

This vignette started from a simple observation: `qvar()` made analytical variance estimation accessible for the most common single-stage case, but the surveys we actually run — and the SAS Poulpe programs we are migrating from — are multistage. `qvar_ms()` extends the `qvar()` philosophy rather than replacing it: the same declarative interface (one argument per methodological feature), the same technical and methodological checks, the same welcome message, and full backward compatibility, since a single-stage call reproduces `qvar()` exactly (case 1). What changes is the reach: an arbitrary number of sampling stages, non-response correction at any subset of stages (declared through response probabilities or conditional weights), calibration at any stage, stratification crossed with the parent units, and a choice of treatments for single-unit strata. The nine cases of this vignette walked through this space one feature at a time, each validated against a hand-built `define_variance_wrapper()` benchmark that makes the underlying Rao (1975) decomposition explicit — the same estimator as the SAS Poulpe macros, which remain the external reference throughout.

Along the way, the wrapper grew from a computing object into a methodological one. The **diagnostics** technical parameter allows a quality control of the variance estimation procedure — variance components with the exact effect of calibration, design effect and its Kish decomposition, degrees of freedom, non-response and domain diagnostics — while `qvar_notes()` and `inspect_wrapper()` make the wrapper itself readable: what happened at definition time, what the encoded design is, and which pipeline will actually execute.

More complex multistage cases — unequal probabilities within strata, balanced sampling at the first stage, shared or integrated weights — are to be handled with the core functions of the **gustave** package, i.e. `define_variance_wrapper()` together with `varDT()`, `var_pois()` and `res_cal()`, following the patterns of the hand-built wrappers of this vignette.

Two directions naturally extend this work. The first is external cross-validation against the `survey` package (Lumley, 2004), the *de facto* standard of the R ecosystem. Its default estimator for multistage designs is the ultimate cluster approximation — collapsing all stages into a with-replacement sampling of primary units — which differs from the stage-by-stage Rao decomposition implemented here: agreement is expected to be close but not exact, and the size and sign of the differences (driven by the first-stage sampling fractions) would be instructive in itself. A future section comparing `qvar_ms()` wrappers with the corresponding `svydesign()` objects, including their respective treatments of lonely PSUs, would give users a second, independent reference point next to `Poulpe`.

The second direction is downstream integration. A `gustave` wrapper cleanly separates the design description from the data, which is powerful for estimation but leaves the pairing of the two to the user. Bundling them into a design object — in the spirit of `survey::svydesign()`, carrying the survey file together with its variance wrapper — would allow `gustave` to plug into the growing ecosystem of tools built on such objects, notably `fonctionr`, a package developed in Belgium that produces publication-ready weighted analyses and visualizations from survey design objects. Precision estimates would then flow directly into graphs and tables, instead of living in a separate results file. This design-object layer is under construction and beyond the scope of this vignette.

6 Annexe

6.1 One responding unit: the Poisson term as the natural continuation of the SRS formula

Two situations must be distinguished in a stratum. When it contains a single *sampled* unit, the SRS variance formula is undefined (the s^2 term divides by $n - 1 = 0$): this is the case handled by the `single_unit_strata` treatment. When it contains several sampled units of which a single one *responds*, the SRS formula remains perfectly defined, provided the non-responding units are kept in the computation with a zero value — which is exactly what the `add_zero()` mechanism of `gustave` does. This appendix shows that, in the latter case, the SRS formula over all n sampled units is **algebraically identical** to the Poisson variance term of the lone respondent. This identity both explains why strata with a single respondent require no special treatment, and motivates the Poisson fallback as the natural continuation of the SRS formula when even the zeros disappear. Notations follow Chauvet’s lecture notes on variance estimation (Ensai): https://ensai.fr/wp-content/uploads/2019/06/TechEchMaster_EstVariance_v1.pdf

6.1.1 From the Horvitz-Thompson variance estimator

Under an arbitrary design, the total is estimated by

$$\hat{t}_{y\pi} = \sum_{k \in S} \frac{y_k}{\pi_k},$$

and its variance is estimated (without bias, provided all joint inclusion probabilities are positive) by

$$\hat{v}_{HT}(\hat{t}_{y\pi}) = \sum_{k, l \in S} \frac{y_k}{\pi_k} \frac{y_l}{\pi_l} \frac{\pi_{kl} - \pi_k \pi_l}{\pi_{kl}}.$$

Suppose a single unit k of the sample carries a non-zero value — say, one respondent with $y_k = 10$ among four units sampled with $\pi = 0.25$, the three non-respondents entering with $y = 0$. Every term of the double sum contains a product $y_k y_l$ and therefore vanishes, except the diagonal term $k = l$. With the convention $\pi_{kk} = \pi_k$, the estimator collapses to

$$\hat{v}_{HT}(\hat{t}_{y\pi}) = \left(\frac{y_k}{\pi_k} \right)^2 \frac{\pi_k - \pi_k^2}{\pi_k} = \left(\frac{y_k}{\pi_k} \right)^2 (1 - \pi_k),$$

which is precisely the contribution of unit k under a Poisson design — the term computed by `var_pois()`.

6.1.2 From the SRS formula with zeroed non-respondents

Under simple random sampling (equal inclusion probabilities $\pi_k = n/N$), the total is estimated by $\hat{t}_{y\pi} = N\bar{y}$ with $\bar{y} = \frac{1}{n} \sum_{k \in S} y_k$, and its variance by the classical formula

$$\hat{v}(\hat{t}_{y\pi}) = N^2 \frac{1-f}{n} s_y^2, \quad s_y^2 = \frac{1}{n-1} \sum_{k \in S} (y_k - \bar{y})^2, \quad f = \frac{n}{N}.$$

With a single non-zero unit k , the sum of squares splits into the contribution of the respondent and that of the $n-1$ zero-valued units, each equal to $\bar{y}^2$:

$$s_y^2 = \frac{1}{n-1} ((y_k - \bar{y})^2 + (n-1)\bar{y}^2) = \frac{1}{n-1} (y_k^2 - 2y_k\bar{y} + n\bar{y}^2).$$

Since $\bar{y} = y_k/n$,

$$s_y^2 = \frac{1}{n-1} \left(y_k^2 - \frac{2y_k^2}{n} + \frac{y_k^2}{n} \right) = \frac{y_k^2}{n-1} \cdot \frac{n-1}{n} = \frac{y_k^2}{n}.$$

Injecting into the full formula, with $f = \pi_k = n/N$:

$$\hat{v}(\hat{t}_{y\pi}) = N^2 \frac{1-f}{n} \frac{y_k^2}{n} = (1 - \pi_k) \frac{N^2 y_k^2}{n^2} = \left(\frac{y_k}{\pi_k} \right)^2 (1 - \pi_k) = \hat{v}_{HT}(\hat{t}_{y\pi}).$$

6.1.3 How to use it in `qvar_ms()` ?

The two computations are rigorously identical: since the Deville-Tillé formula implemented by `varDT()` reduces exactly to the classical SRS estimator under equal inclusion probabilities, the identity holds for the actual `gustave` computation. Two consequences for `qvar_ms()`. First, strata with a single *respondent* among several sampled units require no special treatment: the SRS formula with zeroed non-respondents already computes the right quantity. Second, when the stratum contains a single *sampled* unit, s_y^2 is undefined but the diagonal Horvitz-Thompson term $(1 - \pi_k)(y_k/\pi_k)^2$ remains available: this is the term added by `single_unit_strata = "poisson"`.

7 References

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